Axiom at ArGuard Shared Tasks: Unimodal Text Dominance for Arabic Hate Meme Detection

Md. Ajwad Hossain
Chittagong University of Engineering & Technology
Chittagong, Bangladesh
md.ajwadhossain@gmail.com

Abstract

Detecting hate speech in social media memes requires disentangling complex cross-modal interactions, a challenge magnified in low-resource and morphologically rich languages like Arabic. We present Team Axiom's submission to the ArGuard 2026 Shared Task (Task A1) on binary hate meme classification. While conventional multimodal wisdom posits that text and visual features must be jointly fused, empirical evidence across non-English domains reveals that noisy visual backbones and cross-modal space misalignment frequently degrade language signals. We validate this hypothesis by fine-tuning a unimodal MARBERTv2 sequence classifier without external multimodal fusion. Our simple, well-regularized approach achieved a Macro-F1 of 0.770 and an accuracy of 0.818 on the official gold test set, securing 6th place on the global leaderboard and outperforming both the official AraBERTv2 baseline and our own complex multimodal fusion architectures.

1 Introduction

Multimodal memes combining image macros with embedded text have become a predominant vector for social media engagement, online humor, and toxic content propagation. Detecting hate speech in Arabic memes is particularly demanding due to the prevalence of regional dialects, orthographic variations, sarcasm, and noise introduced during Optical Character Recognition (OCR) (Kmainasi et al., 2026).

The foundational benchmark established by the Hateful Memes Challenge (Kiela et al., 2020) popularized the paradigm that automated systems must rely on deep multimodal fusion to capture juxtaposed text-image semantics. Consequently, recent shared task participants frequently employ complex early, late, or intermediate cross-attention architectures. However, emerging research indicates that unimodal text representations often carry the primary predictive signal in meme classification tasks (Gomez et al., 2020; Karim et al., 2022). In non-English and low-resource settings, standard vision-language models (VLMs) like CLIP fail to capture non-Latin script semantics, causing naive fusion layers to overfit to visual noise (Wagle et al., 2026).

In this paper, we present Team Axiom's system for ArGuard 2026 Task A1 (Alam et al., 2026).[1] Rather than forcing multimodal integration across misaligned vector spaces, we exploit the strong inductive bias of MARBERTv2 (Abdul-Mageed et al., 2021), a language model pre-trained on massive dialectal Twitter corpora. Through rigorous experimentation, we show that our unimodal text-only classifier significantly outperforms our multimodal fusion variants (including Dual Co-Attention and Gated ConvNeXt models), achieving 6th place globally. We substantiate these results with an error analysis revealing the structural limits of text-only classification and the specific failure modes of current multimodal frameworks.

2 Related Work

2.1 Multimodal Fusion Limits & Text Dominance

Although multimodal architectures are theoretically required for memes with benign confounders (Kiela et al., 2020), empirical studies repeatedly demonstrate that unimodal text models frequently match or surpass multimodal fusion systems when visual features are noisy or uninformative. In large-scale benchmarks such as the MMHS150K dataset, multimodal models failed to outperform text-only baselines despite image availability (Gomez et al., 2020). Similarly, evaluations on Bengali meme hate speech datasets

[1] **Content Warning:** This paper contains examples of hate speech, offensive language, and discriminatory imagery from the dataset, which are included solely for illustrative and analytical purposes.

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 400–406
October 28-29, 2026 ©2026 Association for Computational Linguistics

showed that no multimodal fusion model exceeded unimodal text performance, establishing text as the overwhelming carrier of harmful intent (Karim et al., 2022). Further evidence from war meme analysis confirms that visual features primarily assist in entity-targeted hate, whereas text dominates broader hate speech classification (Armenta-Segura et al., 2023). In implicit hate detection, text-only models consistently maintain higher precision over multimodal counterparts (Botelho et al., 2021).

2.2 English-Centric Vision-Language Limitations

A primary driver of multimodal failure in non-English contexts is the structural limitation of pre-trained Vision-Language Models (VLMs). Prominent encoders such as CLIP feature heavily English-centric pre-training distributions, resulting in near-random performance when applied to non-Latin scripts (Wagle et al., 2026). Rather than learning genuine cross-modal semantics or culturally specific iconography, CLIP-based encoders rely on visual artifacts like text density and typographic weight (Wagle et al., 2026). Furthermore, dual-stream VLMs struggle with fine-grained visual perception on unseen classes (Liu et al., 2024) and lack robust cross-lingual alignment without extensive multilingual pre-training (Han et al., 2025).

2.3 Visual Noise and Fusion Failures

Static fusion mechanisms that concatenate independently extracted text and image vectors fail to model dynamic cross-modal interactions (Anaissi et al., 2025). Under conditions of limited training data, standard ensembling and naive fusion violate theoretical independence assumptions, leading to correlated overfitting that actively degrades classification accuracy compared to constrained unimodal baselines (Wagle et al., 2026). While structured intermediate cross-attention and learnable gating can partially mitigate visual noise by dynamically weighting modalities, the cross-modal representation gap in Arabic meme detection often renders pure linguistic representation fine-tuning more resilient.

2.4 Arabic Pre-trained Language Models

Transformer-based language models pre-trained on native Arabic text have advanced dialectal NLP tasks. Models such as MARBERTv2 (Abdul-Mageed et al., 2021), pre-trained on 1 billion Arabic tweets, offer exceptional coverage of informal social media register, regional slang, and character elongation. This robust linguistic baseline enables sequence classifiers to capture implicit sarcasm and offensive dialectal phrasing from raw OCR outputs without requiring aggressive text normalization or fragile multimodal alignment layers.

3 Methodology

3.1 Data Processing

We utilized the official splits provided by the organizers (Kmainasi et al., 2026). Unlike approaches that heavily sanitize input text, we applied minimal preprocessing. Arabic memes frequently utilize dialectal slang, emojis, and elongated characters to convey sarcasm or aggression. Aggressive normalization strips these critical semantic markers; thus, the raw OCR-extracted text was fed directly into the tokenizer.

3.2 Unimodal Text Architecture (Submitted)

Our primary and best system relies exclusively on the text modality. We employed MARBERTv2 (Abdul-Mageed et al., 2021), a bidirectional Transformer model ideally suited for this domain. We attached a standard linear sequence classification head over the pooled output of the `[CLS]` token. The model was trained using standard Cross-Entropy Loss, explicitly avoiding complex class-weighting or focal loss, as our exploratory data analysis revealed a manageable class distribution that did not necessitate aggressive minority-class scaling.

3.3 Multimodal Architectures (Comparative Baselines)

To validate our unimodal hypothesis, we developed and evaluated three distinct multimodal architectures:

- **Naive Fusion (Concatenation):** A baseline early-fusion model concatenating the `[CLS]` embedding of MARBERTv2 with the global average pooled features of a frozen CLIP-ViT model.
- **Dual Co-Attention:** A symmetric cross-attention architecture using Arabic-CLIP, allowing text tokens to query visual patches and vice versa before mean pooling.

- **Gated ConvNeXt:** A residual-gated network where MARBERTv2 serves as the primary anchor, and local spatial features from a ConvNeXt-Tiny visual backbone are injected via a learned non-linear sigmoid gate.

4 Experimental Setup

Our final MARBERTv2 model was implemented using PyTorch and the Hugging Face `transformers` library. The text was tokenized with a maximum sequence length of 128. We fine-tuned the model for 20 epochs using the AdamW optimizer with a learning rate of 3×10^{-5} and a batch size of 16. All experiments were conducted on a single NVIDIA GPU using FP16 mixed precision.

Data Splits & Evaluation: The AHA-Memes dataset was partitioned by the organizers into 3,500 training samples, 500 development samples, and 500 unlabelled test samples. We strictly maintained these original splits and did not merge the development data into the training pool. The experimental pipeline proceeded as follows:

1. **Training phase:** The model weights were updated exclusively using the 3,500 training samples.

2. **Validation phase:** At the end of each epoch, the model was evaluated on the isolated 500-sample development set. We monitored the development Macro-F1 score and utilized strict checkpointing to save only the model state that yielded the peak validation performance, preventing over-training.

3. **Testing phase:** The top-performing checkpoints were utilized to generate predictions on the unlabelled test set. To obtain final metrics without iteratively overfitting to the leaderboard, we submitted our specific candidate models (our primary text-only model and five multimodal comparative baselines) to the official CodaBench evaluation server.

5 Results and Discussion

5.1 Quantitative Results

Table 1 presents the comparative results of our different system architectures against the official test set metrics.

The empirical data strongly supports our core hypothesis. The Vision-Only baseline achieved a mere 0.625 F1, indicating that images alone rarely contain explicit hate without textual context. Furthermore, Naive Fusion (0.648) and Dual Co-Attention (0.689) performed worse than advanced unimodal models. While sophisticated gating mechanisms (Gated ConvNeXt, 0.744) helped filter out visual noise, they still failed to surpass the elegantly simple MARBERTv2 text-only baseline (0.770).

To address the precise source of our unimodal model's advantage, we conducted a class-wise evaluation (detailed in Appendix C). The analysis reveals that while complex multimodal fusion marginally improved the recall of the hateful class, it suffered severe precision drops due to visual noise, aggressively hallucinating hate from benign images. Conversely, our text-only MARBERTv2 model maintained a superior hateful-class precision (0.732) and dominated the safe majority class with a precision of 0.846 and recall of 0.906.

5.2 Error Analysis

To understand the limitations of our unimodal approach, we conducted a qualitative and quantitative analysis of the 91 misclassified instances (33 False Positives, 58 False Negatives) using the released gold test labels. To isolate vocabulary biases, we applied a TF-IDF analysis to the texts of these misclassified samples. A detailed breakdown of specific memes is provided alongside visual examples in Appendix B.

False Positives (Lexical and Geopolitical Bias): Our model occasionally over-indexed on highly charged vocabulary, missing the surrounding comedic context. TF-IDF analysis revealed that alongside common stopwords, terms like علم (flag) and عاجل (urgent/breaking) were highly weighted in false positive errors. Without visual context, the text-only classifier struggles to distinguish between the benign mention of a sensitive political topic and actual hateful intent.

False Negatives (Benign Masking & Visual Dependency): Conversely, the 58 false negatives revealed scenarios where the hateful signal was either entirely reliant on the image or masked by mundane text. TF-IDF extraction highlighted everyday terms like البنات (girls) and الشغل (work) as primary drivers of these errors. Because this vocabulary is statistically safe in the training distribution, the model defaults to a non-hateful predic-

Model Configuration	Modality	Macro-Precision	Macro-Recall	Accuracy	Macro-F1
CLIP-ViT (Zero-Shot)	Vision-Only	0.662	0.617	0.726	0.626
Naive Fusion (Concat)	Multimodal	0.677	0.638	0.736	0.648
Official AraBERTv2	Text-Only	–	–	0.706	0.678
Dual Co-Attention	Multimodal	0.684	0.709	0.722	0.690
Gated ConvNeXt	Multimodal	0.741	0.749	0.784	0.745
Residual Gated Fusion	Multimodal	0.746	0.775	0.782	0.755
MARBERTv2 (Final)	**Text-Only**	**0.789**	**0.757**	**0.818**	**0.770**

Table 1: Test set performance metrics across different architectures. Our unimodal text model outperformed all multimodal fusion attempts across precision, recall, and overall F1.

tion, completely missing visual caricatures or orthographic evasion tactics (e.g., elongating vulgar slang to bypass filters) that carry the true hateful signal.

6 Conclusion

In this paper, we detailed Team Axiom's submission to the ArGuard 2026 Shared Task. Through a systematic comparison of unimodal and multimodal architectures, we demonstrated that the linguistic signal in Arabic hate memes overwhelmingly dominates the visual signal. Our highly optimized, text-only MARBERTv2 model bypassed the gradient shock and overfitting issues common to multimodal fusion, achieving 6th place on the official leaderboard. Future work will explore advanced Vision-Language Models natively pretrained on Arabic text-image pairs to better resolve visual-dependent irony without sacrificing the core linguistic accuracy.

Limitations

While our unimodal MARBERTv2 classifier demonstrates robust performance, relying exclusively on text introduces structural limitations. As highlighted in our error analysis, the model inherently fails when hateful intent is conveyed solely through visual irony or when benign text is juxtaposed with discriminatory imagery. Furthermore, intentional orthographic evasion techniques, such as extreme character elongation, can force critical tokens out of the subword vocabulary, masking the offensive signal. Additionally, memes frequently rely on subtle geopolitical, historical, or cultural referencing that requires external world knowledge to decode. Because our system operates without access to an external or internal knowledge base, it struggles to recognize highly contextual hate speech that relies on these subtle real-world references. Finally, the system's performance is intrinsically tied to the linguistic distribution of the MARBERTv2 pre-training corpus and may not generalize optimally to emerging dialects, novel regional slang, or heavy code-switching not represented in the training data.

Ethical Considerations

Automated hate speech detection carries the risk of false positives, where benign commentary or irony is misclassified as hateful, potentially suppressing legitimate free expression. Therefore, our system is designed to assist human moderators rather than serve as an autonomous censorship tool. Additionally, while the AHA-Memes dataset contains highly offensive content, our engagement was strictly limited to analytical purposes for improving online safety, and we did not generate or propagate any synthetic toxic material during this study.

Acknowledgements

The author would like to thank the organizers of the ArGuard 2026 Shared Task and the creators of the AHA-Memes dataset for providing the resources and platform to advance Arabic safety research.

References

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. ARBERT & MARBERT: Deep bidirectional transformers for Arabic. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7088--7105, Online. Association for Computational Linguistics.

Firoj Alam, Md. Rafiul Biswas, Mohamed Bayan

Kmainasi, Ali Ezzat Shahroor, Hamdy Mubarak, Georgios Mikros, Abul Hasnat, and Wajdi Zaghouani. 2026. ArGuard Shared Task: Harmful content detection in Arabic Memes and LLM Prompts. In *Proceedings of the Fourth Arabic Natural Language Processing Conference: Shared Tasks*, Budapest, Hungary. Association for Computational Linguistics.

Ali Anaissi, Junaid Akram, Kunal Chaturvedi, and Ali Braytee. 2025. Detecting and understanding hateful contents in memes through captioning and visual question-answering. page 134146.

Jesus Armenta-Segura, César Jesús Núñez-Prado, Grigori Olegovich Sidorov, Alexander Gelbukh, and Rodrigo Francisco Román-Godínez. 2023. Ometeotl@multimodal hate speech event detection 2023: Hate speech and text-image correlation detection in real life memes using pretrained BERT models over text. In *Proceedings of the 6th Workshop on Challenges and Applications of Automated Extraction of Socio-political Events from Text*, pages 53--59, Varna, Bulgaria. INCOMA Ltd., Shoumen, Bulgaria.

Austin Botelho, Bertie Vidgen, and Scott A. Hale. 2021. Deciphering implicit hate: Evaluating automated detection algorithms for multimodal hate. *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 1896--1907.

Raul Gomez, Jaume Gibert, Lluis Gomez, and Dimosthenis Karatzas. 2020. Exploring hate speech detection in multimodal publications. In *2020 IEEE Winter Conference on Applications of Computer Vision (WACV)*, pages 1459--1467.

Xiaofeng Han, Shunpeng Chen, Zenghuang Fu, Zhe Feng, Lue Fan, Dong An, Changwei Wang, Li Guo, Weiliang Meng, Xiaopeng Zhang, et al. 2025. Multimodal fusion and vision-language models: A survey for robot vision. *Information Fusion*, page 103652.

Md Rezaul Karim, Sumon Kanti Dey, Tanhim Islam, Md Shajalal, and Bharathi Raja Chakravarthi. 2022. Multimodal hate speech detection from bengali memes and texts. In *International Conference on Speech and Language Technologies for Low-resource Languages*, pages 293--308. Springer.

Douwe Kiela, Hamed Firooz, Aravind Mohan, Vedanuj Goswami, Amanpreet Singh, Pratik Ringshia, and Davide Testuggine. 2020. The hateful memes challenge: Detecting hate speech in multimodal memes. *Advances in neural information processing systems*, 33:2611--2624.

Mohamed Bayan Kmainasi, Ali Ezzat Shahroor, Abul Hasnat, Md. Rafiul Biswas, Wajdi Zaghouani, and Firoj Alam. 2026. Aha-memes: A fine-grained multimodal benchmark for understanding hate in arabic memes. *Preprint*, arXiv:2607.27393.

Xin Liu, Jiamin Wu, Wenfei Yang, Xu Zhou, and Tianzhu Zhang. 2024. Multi-modal attribute prompting for vision-language models. *IEEE Transactions on Circuits and Systems for Video Technology*, 34(11):11579--11591.

Samir Wagle, Reewaj Khanal, and Abiral Adhikari. 2026. Meme-fusion@chipsal 2026: Multimodal ablation study of hate detection and sentiment analysis on nepali memes. *Preprint*, arXiv:2604.14218.

A Dataset Statistics

The ArGuard Task 1 (AHA-Memes) dataset was partitioned into Train, Development, and Test splits. The training data contained 3,500 samples (37.83% Hateful, 62.17% Not Hateful). The development set comprised 500 samples (37.80% Hateful, 62.20% Not Hateful), and the final unlabelled test set contained 500 samples. Notably, the class imbalance ($\sim$62/38) was moderate enough that our system achieved optimal convergence using standard Cross-Entropy loss, without requiring complex focal scaling or synthetic data augmentation.

B Qualitative Error Examples

To illustrate the phenomenon of visual dependency and lexical trigger bias discussed in Section 5.2, we present representative examples from our error analysis.

B.1 False Negatives (Visual Dependency)

In these examples, the text-only model incorrectly predicted "Not Hateful" because the hateful intent relied on visual irony or masked text.

False Negative (Visual Sarcasm)
Gold: Hateful | Predicted: Not Hateful



Figure 1: Visual Dependency on Historical Irony: The text combines dark aviation commentary with a reference to a viral Egyptian livestream persona (*"Let me just strike one more tower"*). To the text-only model, this reads as an ambiguous piece of dark humor; the culturally specific satirical intent relies entirely on the visual pairing.

False Negative (Visual Sarcasm)
Gold: Hateful | Predicted: Not Hateful



Figure 2: Targeted Regional Stereotyping: The text lists contrasting subculture archetypes paired with specific Cairo neighborhoods (*"Goth girl from Al-Attaba, Femboy from Maadi"*). To the text model, these appear as benign geographic and pop-culture references. The underlying classist or regional satire relies entirely on the visual contrast between the traditional commercial district archetype and the upscale cosmopolitan stereotype.

B.2 False Positives (Lexical Trigger Bias)

In these examples, the text-only model incorrectly predicted "Hateful" because it over-indexed on strong vocabulary, missing the comedic visual context.

False Positive (Lexical Bias)
Gold: Not Hateful | Predicted: Hateful



Figure 3: Geopolitical Bias: The model flagged this due to the co-occurrence of geopolitical trigger words like "burning," "flag," and "Israel." The image context reveals it is a benign meme mocking a geographic mistake (burning the Argentinian flag instead).

False Positive (Lexical Bias)
Gold: Not Hateful | Predicted: Hateful



Figure 4: Economic Humor Bias: The model flagged this text as hateful, likely over-indexing on the dramatic phrasing of ancestors watching a bloodline end. In reality, it is a benign, self-deprecating joke about youth financial struggles and the high cost of marriage.

C Class-Wise Performance Breakdown

To further isolate the effects of multimodal fusion versus unimodal text classification, Table 2 presents the detailed class-wise Precision, Recall, and F1 scores extracted from the official CodaBench evaluation logs.

Model	Hateful			Not Hateful		
	P	**R**	**F1**	**P**	**R**	**F1**
CLIP-ViT (Zero-Shot)	0.559	0.351	0.432	0.764	0.884	0.819
Naive Fusion	0.578	0.399	0.472	0.776	0.878	0.824
Dual Co-Attention	0.524	0.676	0.590	0.845	0.741	0.790
Gated ConvNeXt	0.628	0.662	0.645	0.855	0.835	0.845
Residual Gated	0.605	**0.757**	**0.673**	**0.886**	0.793	0.837
MARBERTv2 (Final)	**0.732**	0.608	0.664	0.846	**0.906**	**0.875**

Table 2: Class-wise performance breakdown. P = Precision, R = Recall. The fusion models (e.g., Residual Gated) achieve high hateful recall but suffer from poor precision. Our unimodal text model (MARBERTv2) balances high precision on the hateful class with dominant performance on the benign majority class.

The breakdown clearly illustrates the "visual noise" phenomenon. While the Residual Gated Fusion model successfully flags a high volume of hateful memes (Recall: 0.757), its precision drops significantly (0.605), indicating a high rate of false positives triggered by benign visual confounders. Our unimodal text baseline avoids this cross-modal gradient shock, securing the highest precision on the hateful class (0.732) and ensuring robust, reliable classification across the dataset.